

demonstrating that age is no barrier when using the right tools.

Once the basics of programming and hardware are in place, machine learning can be integrated into projects. Utilise open source resources and attend tech events—even virtually. Platforms such as Electronics For You offer opportunities to learn and collaborate.

Platforms like IEEE, GitHub, Medium, and Dev offer a space to publish your work and contribute to the ecosystem. Some companies have excelled simply by leveraging open source projects and datasets to train models.

Avoid common pitfalls, which include trying to learn everything at once, avoiding hardware, or isolating yourself. Use simulators to learn hardware logic if buying equipment is a hurdle. Connect with others; in person, not just online. Sharing ideas helps align mindsets and encourages collective growth.

Then comes the role of generative AI. After building simple projects, explore tools that automate code documentation or generate test scripts. This helps address common pain points in software development. Documentation, for example, is often neglected, but generative AI can help create and maintain it efficiently.

Let me narrate another story. Ten years ago, I hired three talented engineers who jumped straight into coding. Two years later, they could not recall what the code did because there was no documentation. Tools today can fill such gaps. Use them judiciously, whether it is generating drivers or automating test cases.

Coming to career opportunities, globally, countries like the US, Germany, India, Canada, and Singapore are hiring in AI and IoT. With remote work becoming the norm, engineers from India are working for global companies without needing to relocate. This opens vast opportunities; something that was not possible even a decade ago.

Also, reflect regularly. In agile methodology, we conduct retrospectives every two weeks. Apply the same to your learning; review what you have learned every six months or year. Continuous learning is vital.

Moreover, explore AI frameworks such as TensorFlow and understand deep learning architectures, including CNNs and RNNs. Decide which area suits you best: whether it is speech recognition, natural language processing (NLP), or medical imaging, and focus your learning accordingly.

When we talk about embedded systems and edge computing, micro-

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controllers and DSPs are central. Tools like Raspberry Pi, Arduino, and Nvidia Jetson play a key role. Jetson, in particular, is a powerful platform that I highly recommend. Beginners should start with simulators before choosing hardware, keeping in mind the power constraints of battery-operated devices.

Control systems and robotics are also critical areas. Reinforcement learning is widely used in robotics because it learns through feedback, making it a strong and adaptable approach.

NLP is another vital domain. Applications such as speech-to-text, text-to-speech, and voice assistants like Alexa and Google Home are everyday examples of AI in action.

In the realm of IoT and AI, predictive maintenance stands out

as a key technology. Unlike reactive maintenance (which occurs after failure) or preventive maintenance (which aims to avoid failure), predictive maintenance anticipates failure before it occurs, utilising AI-driven analytics, which are vital for smart devices.

Sensor fusion and AI-enabled data analytics help make sense of this data. Tools like Power BI, Tableau, and Amazon QuickSight can visualise large data sets. Learning these tools is often the first step toward actionable insights.

Ethical AI and governance are becoming increasingly important. Safety is paramount. A recent European conference addressed AI misuse, leading some countries to temporarily restrict tools like ChatGPT over safety concerns. Now, global AI governance frameworks are taking shape.

Gartner outlines a strategic AI framework: AI strategy, value, organisation, people and culture, governance, engineering, and data. These elements include defining vision, prioritising use cases, measuring maturity, preparing a skilled workforce, and making decisions like building versus buying models. Data readiness is also essential; assessing and preparing data is foundational to successful AI deployment.

Understanding this broader organisational context can help engineers align with strategic goals. Many companies are upskilling existing employees to prepare for AI-driven roles.

Ultimately, innovation is key. We cannot rely solely on tools built by others. Every engineer should strive to create something new, whether in the realm of IoT, embedded systems, or AI.

Looking at the future...

Today's teams already include roles like Scrum Master, DevOps Engineer, Product Manager, and Embedded Specialist. But AI is introducing new roles: Prompt Engineers, for example,